

Co-Expression of PdxJ Increases Soluble EnzQ Production Without Detectable Binding to Purified EnzQ

Lena Sato, Quinn Nyx, and Omar Bren | Model Protein Interaction Facility

ABSTRACT

Co-expression of the fictional folding factor PdxJ increased soluble EnzQ from 2.4 to 7.1 mg/L culture. To determine whether PdxJ remained bound to EnzQ, purified proteins were tested by nickel pull-down, size-exclusion chromatography, and microscale thermophoresis. No stable complex was detected under the tested conditions, with an estimated lower limit of detectable affinity near 50 μ M. The results support an effect of PdxJ during expression but not persistent binding after purification. A transient or indirect mechanism is possible but was not demonstrated.

Keywords: *co-expression, soluble expression, pull-down, no binding, EnzQ*

INTRODUCTION

Co-expression partners can improve recombinant protein solubility through direct chaperoning, metabolic effects, or altered translation. Increased soluble yield alone does not prove a stable physical complex.

This study separated the production question from the binding question. First, EnzQ was expressed with or without PdxJ. Second, purified proteins were mixed and tested by three orthogonal assays.

MATERIALS AND METHODS

Co-expression

His8-EnzQ was expressed alone or with untagged PdxJ at 20 C for 16 h. Soluble EnzQ was purified by nickel affinity chromatography and quantified after correction for purity.

Binding assays

For pull-down experiments, purified EnzQ and PdxJ were mixed at a 1:2 molar ratio before nickel capture. Size-exclusion chromatography used a calibrated analytical column. Microscale thermophoresis tested PdxJ from 0.05 to 50 uM against fluorescent EnzQ.

Interpretation threshold

A stable interaction required concordant evidence from at least two assays. A weak trend in one assay was not treated as binding.

Assay	Readout	Positive criterion
Ni pull-down	PdxJ co-elution	Above background
SEC	Peak shift/new peak	Reproducible shift
MST	Binding curve	Finite Kd within range

Table 1. Stable-binding criteria.

RESULTS

PdxJ increased soluble EnzQ

EnzQ expressed alone yielded 2.4 +/- 0.5 mg/L soluble protein. Co-expression with PdxJ increased the yield to 7.1 +/- 0.8 mg/L, a 3.0-fold increase. Total EnzQ expression increased only 1.2-fold, indicating redistribution toward the soluble fraction.

No stable complex was detected

PdxJ did not co-elute above background in the nickel pull-down. Mixing purified proteins did not shift the EnzQ size-exclusion peak. Thermophoresis signals remained flat up to 50 uM PdxJ and could not be fit to a binding isotherm.

What the experiment does not show

The assays do not exclude a short-lived interaction during translation or an indirect effect on the expression host. They do show that purified EnzQ and PdxJ did not form a stable complex under the reported buffer conditions.

Outcome	EnzQ alone	EnzQ + PdxJ
Soluble yield (mg/L)	2.4 +/- 0.5	7.1 +/- 0.8
Total expression (relative)	1.00	1.20
Stable binding detected	No	No

Table 2. Co-expression increased soluble yield without evidence of stable binding.



Figure 1. PdxJ co-expression increased the amount of soluble EnzQ.

DISCUSSION

PdxJ improved soluble EnzQ production substantially, but no stable PdxJ-EnzQ complex was detected after purification. The most conservative interpretation is that PdxJ acts during expression or indirectly through the host.

A transient interaction is biologically plausible and is shown in an illustrative model in the laboratory notebook, but plausibility is not evidence. The present paper does not demonstrate direct binding.

LIMITATIONS

The binding assays were performed in one buffer and may miss interactions weaker than approximately 50 μ M or interactions requiring additional cellular factors.

CONCLUSION

PdxJ co-expression increased soluble EnzQ yield threefold, but purified PdxJ and EnzQ showed no detectable stable interaction in pull-down, size-exclusion, or thermophoresis assays.

REFERENCES

The following references are fictional and are included only to test reference-section handling.

1. Vale M, Karst T. Quantitative controls for co-expression and protein solubility. *Journal of Synthetic Methods* 12:101-109 (2018).
2. Orin P, Sato L. Reproducible analysis of co-expression and protein solubility in microbial systems. *Bench Biology* 7:44-53 (2019).
3. Kess D, Miro H. Sources of variance during co-expression and protein solubility. *Experimental Simulation Reports* 4:1-12 (2020).
4. Nyx R, Bell A. Statistical treatment of small-scale co-expression and protein solubility studies. *Fictional Biochemistry* 9:77-86 (2021).
5. Fenn J, Ilya V. Buffer composition and apparent recovery in co-expression and protein solubility. *Model Process Journal* 15:203-214 (2022).
6. Rao N, Elsen C. Practical controls for affinity-based co-expression and protein solubility. *Synthetic Protein Science* 2:55-64 (2022).
7. Wen Q, Daro S. Avoiding over-interpretation in co-expression and protein solubility. *Journal of Benchmark Research* 6:90-99 (2023).
8. Pike E, Lorn B. Data provenance for co-expression and protein solubility. *Simulated Laboratory Notes* 11:18-29 (2023).
9. Yara G, Moss K. Negative results in co-expression and protein solubility. *Null Findings Quarterly* 3:5-14 (2024).
10. Bren U, Cole M. Reporting standards for co-expression and protein solubility. *Benchmark Standards* 1:1-8 (2025).